Imagine a tool that's powerful enough to drive artificial intelligence, easy enough to create websites, flexible enough to analyze data, and handy for scientific research—all while being beginner-friendly! That's **Python**.

Total Duration: approx**. 30 Hours - Theory + Coding (Practical)**

That includes:

- 10+ Basic Projects
- 3 Portfolio Projects
- Career Guidance

---

print("Welcome to Python ! Your journey into code begins here.")

Course Outline

---

1. Introduction to Python Programming (2 hours)

Theory (1.5 hours)

- Overview of Python and its applications
- Setting up the environment (Anaconda, Jupyter Notebooks, VSCode)
- Python Syntax, Variables, and Data Types

Coding (0.5 hours)

2. Control Structures and Functions (3 hours)

Theory (1.5 hours)

- Conditional statements: if, else, elif
- Loops: for, while, and control keywords (break, continue)
- Functions: Defining, calling, return values, lambda functions
- Scope and Lifetime of Variables

Coding (1.5 hours)

3. Data Structures (3 hours)

Theory (2 hours)

- Lists, Tuples, Sets, Dictionaries
- List and dictionary comprehensions
- Basic operations and methods for each data structure
- Common use cases in data manipulation

Coding (1 hour)

4. Object-oriented Programming (OOP) (4 hours)

Theory (3 hours)

- Classes and Objects
- Constructors, Destructors, and self keyword
- Inheritance, Polymorphism, Encapsulation, and Abstraction
- SOLID Principles

Coding (1 hour)

5. Error Handling and Debugging (2 hours)

Theory (1.5 hours)

- Types of errors in Python
- Exception handling: try, except, else, finally
- Debugging techniques and using logging

Coding (0.5 hours)

6. File Handling (2 hours)

Theory (1.5 hours)

- Reading and writing files (open, read, write, close)
- Working with JSON and CSV files, etc
- Context managers and ‘with’ statements

Coding (0.5 hours)

7. Introduction to Modules and Libraries (3 hours)

Theory (2 hours)

- Standard libraries (math, datetime, os, etc.)
- Third-party libraries (overview of numpy, pandas, matplotlib)
- Creating and importing custom modules

Coding (1 hour)

8. Data Science and Visualization Basics (4 hours)

Theory (3 hours)

- Introduction to Data Science with Python
- Using pandas for data manipulation
- Visualization techniques with Matplotlib and Seaborn

Coding (1 hour)

9. Introduction to Machine Learning (3 hours)

Theory (2 hours)

- Overview of machine learning applications and trends

- Introduction to scikit-learn and machine learning workflow
- Simple ML algorithms (e.g., linear regression, decision trees)

Coding (1 hour)

10. Advanced Python Concepts and Practices (2 hours)

Theory and Review (1 hour)

- Decorators and Generators
- Advanced Data Structures (Stacks, queues, linked lists, and their use cases.)
- Multithreading and Multiprocessing

Coding (1 hour)

Prepared by: Sujan Sharma (ML Engineer)